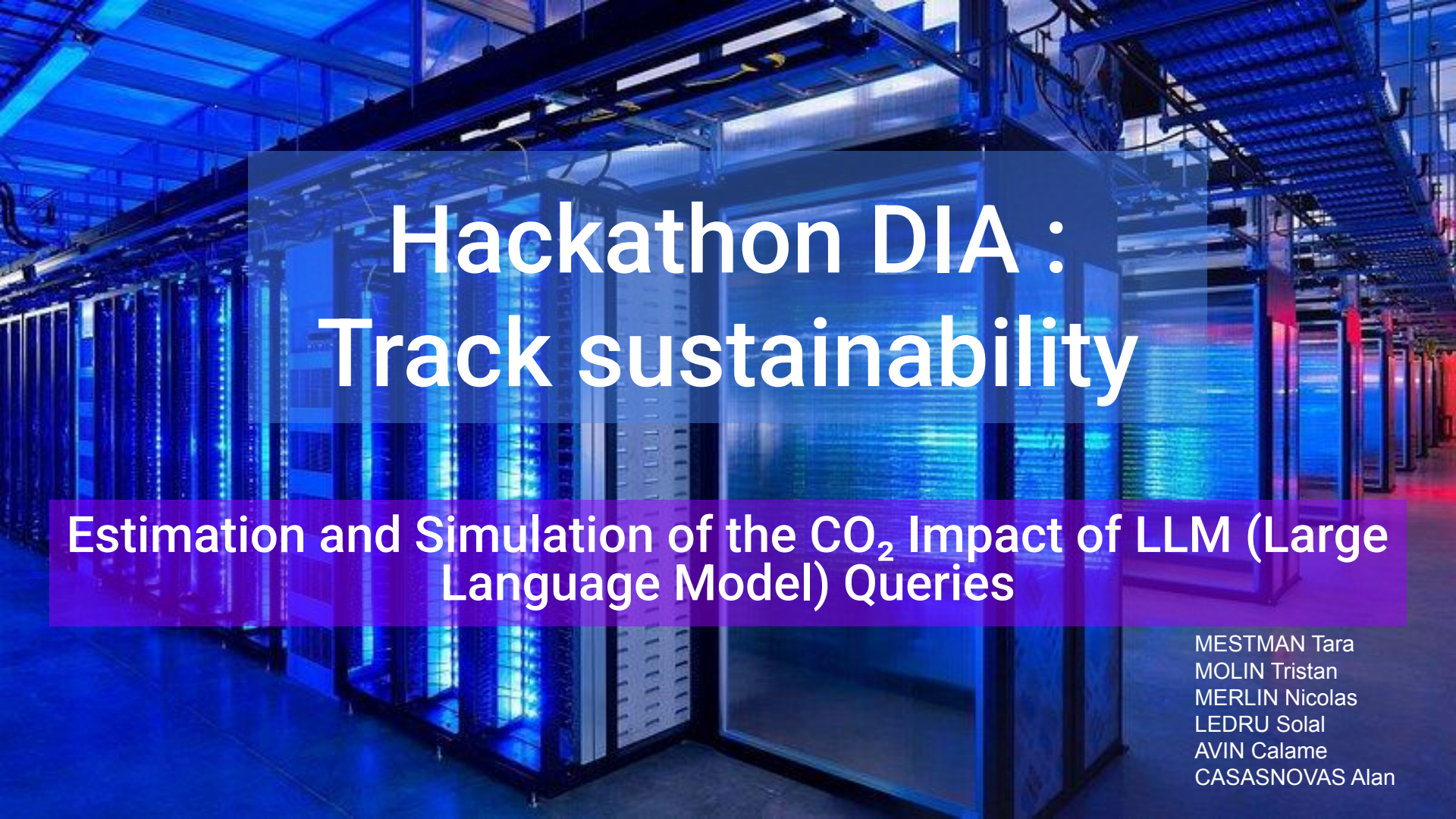




Hackathon DIA : Track sustainability

Estimation and Simulation of the CO₂ Impact of LLM (Large Language Model) Queries

MESTMAN Tara
MOLIN Tristan
MERLIN Nicolas
LEDRU Solal
AVIN Calame
CASASNOVAS Alan

Our use case

Predict the energy consumption and CO² emissions of a request to a LLM model

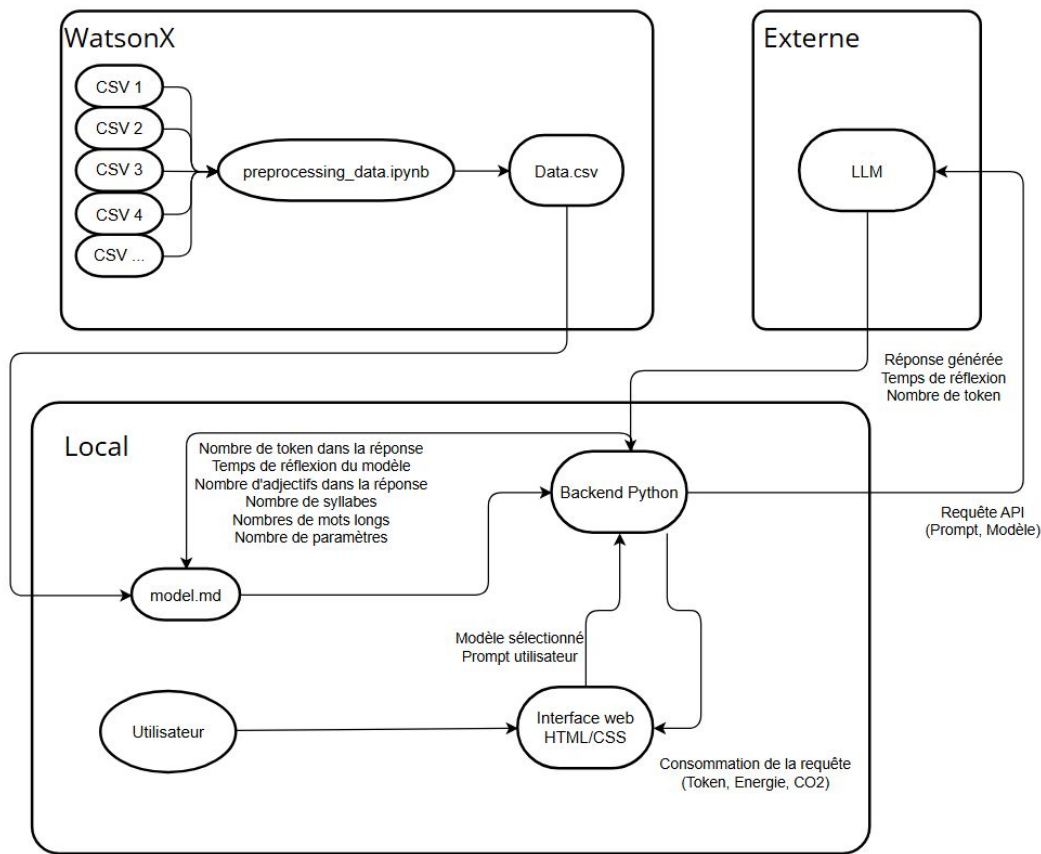
Actifs de données		
☐	 	codefeedback_gemma_7b_workstation.csv
☐	 	codefeedback_codellama_7b_laptop1.csv
☐	 	codefeedback_gemma_2b_laptop2.csv
☐	 	alpaca_gemma_2b_workstation.csv
☐	 	alpaca_gemma_7b_workstation.csv
☐	 	codefeedback_codellama_7b_workstation.csv
☐	 	codefeedback_gemma_7b_laptop2.csv
☐	 	codefeedback_gemma_2b_workstation.csv
☐	 	alpaca_gemma_2b_laptop1.csv
☐	 	codefeedback_codellama_70b_workstation.csv
☐	 	alpaca_llama3_8b_laptop2.csv
☐	 	alpaca_gemma_7b_laptop2.csv
☐	 	codefeedback_codellama_7b_laptop2.csv
☐	 	alpaca_llama3_70b_server.csv
☐	 	alpaca_gemma_2b_laptop2.csv

1. Train a ML model on datasets of prompt requests and answers to predict the energy consumption (kWH) of LLM models
2. Convert the energy consumption into CO² emissions (g)
3. Build an interactive user interface to request a variety of LLM and compare their energy consumption

Why ?

- Shows LLM users the impact of their usage
- Incite them to use LLM intelligently
- Aiming to reduce LLM environmental impact

Structure of our project



Data preprocessing

We merged all datasets together, cleaned the fused dataset by removing empty rows and useless or redundant columns.

We created a new feature representing the number of parameters of each model : **model_size**.

We selected the features that had a correlation of at least 0.09 with our target feature.

```
'Unnamed: 0',          #Index automatique pandas
'index',               #Index redondant
'created_at',          #Timestamp de création (sans intérêt)
'prompt',              #Texte du prompt (trop volumineux)
'response',            #Texte de la réponse (trop volumineux)
'type',                #Type de donnée (sans intérêt)
'clock_duration',      #Durée (redondant avec d'autres métriques)
'start_time',          #Heure de début (sans intérêt)
'end_time',            #Heure de fin (sans intérêt)
'energy_consumption_llm', #Consommation partielle (on garde le total)
'energy_consumption_monitoring', #Consommation du monitoring
'energy_consumption_llm_cpu', #Consommation CPU seule (On choisit le total parce que :)
'energy_consumption_llm_gpu', #Consommation GPU seule (Le hardware n'a pas toujours de GPU)
'adjectives',          #Nombre d'adjectifs (redondant)
'adverbs',             #Nombre d'adverbes (redondant)
```

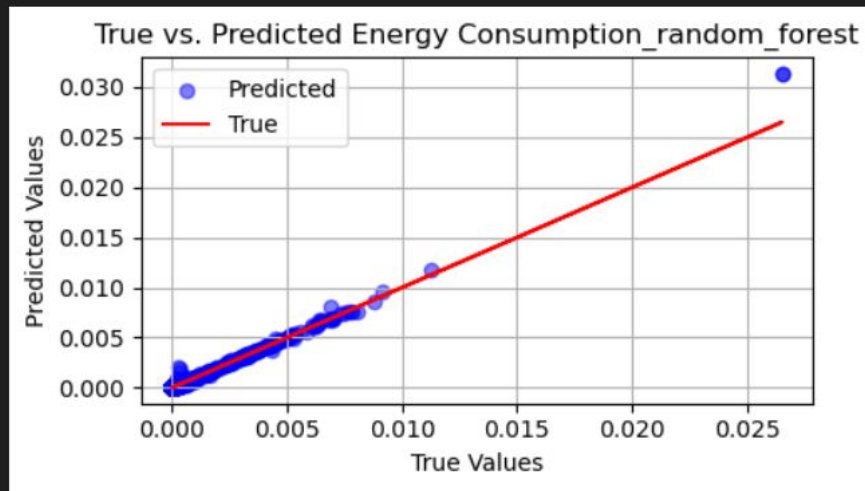
```
Corrélations avec 'energy_consumption_llm_total' :
energy_consumption_llm_total      : 1.0000
response_token_length             : 0.8456
response_duration                 : 0.6248
total_duration                   : 0.6177
model_size                       : 0.3821
adj_count                        : 0.1129
polysyllabcount                  : 0.0945
long_word_count                   : 0.0904
syllable_count                   : 0.0832
letter_count                     : 0.0822
char_count                      : 0.0804
reading_time                    : 0.0804
lexicon_count                   : 0.0776
prompt_token_length              : 0.0755
adverb_count                    : 0.0731
stop_word_count                 : 0.0721
noun_count                      : 0.0698
monosyllabcount                 : 0.0684
word_count                      : 0.0679
verb_count                      : 0.0668
```

Model training

We tested multiple models, RandomForestRegressor stood out as the optimal model for our use case.

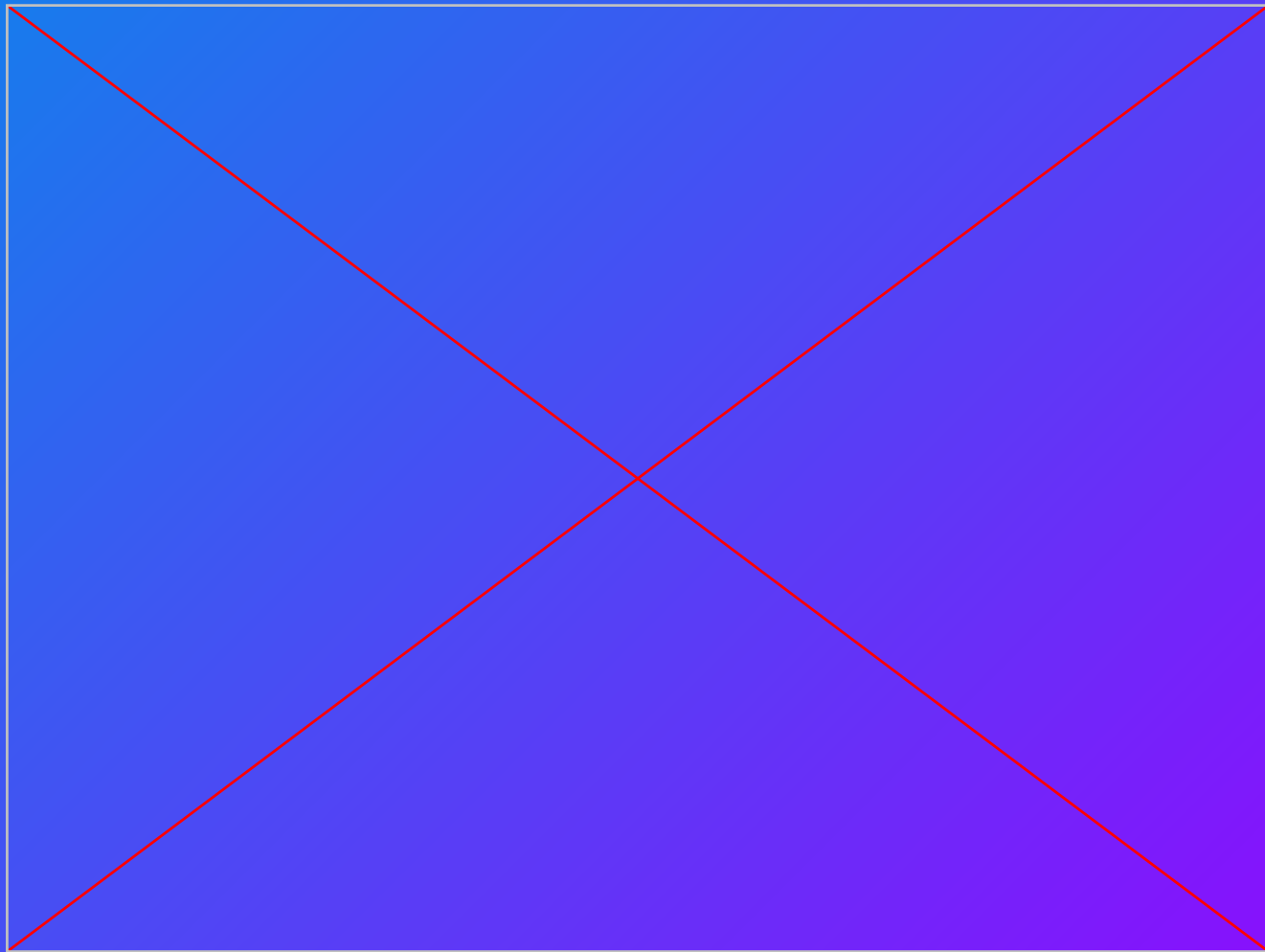
```
models = {  
    "random_forest": RandomForestRegressor(n_estimators=100, random_state=42, max_depth=15, n_jobs=-1),  
    "decision_tree": DecisionTreeRegressor(),  
    "xgboost": xgb.XGBRegressor(),  
    "linear_regression": LinearRegression(),  
}
```

mse: 4.039415791317722e-09 & mae: 6.896921413127833e-06 & r²: 0.9849131513059202



The model achieved a R² score of 0.985, confirming the findings of the researchers that made the dataset.

Demo



How our solution works

Presenting the user to the consumption of their prompts.

Collecting data from LLM APIs, we apply a Random Forest Regressor to evaluate the consumption of every prompt and answer between the user and LLMs.

This will help LLM users understand the direct correlation between their day to day interactions with AI and their impacts on the planet.

We want to encourage every major chatbot platform to do the same : showing transparency will better the AI world.

Into the future, how can we go further?

- Comparison of different models, showing which one is the least energy-hungry
- Showing an equivalent of the distance travelled by car
- Give tips on how to reduce the consumption

- Improve the interface
- Make it easier to switch between models
- Improve the appearance of the output

